

Roki Roy

Backend Developer

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Summary

Backend-focused engineer with production experience building Go-based REST APIs, authentication systems, and scalable backend services. Capable of delivering full-stack solutions using React, Next.js, and Flutter. Strong foundation in clean architecture, database design, API integration, and Docker-based deployment.

Experience

Software Development Engineer (SDE 1) — Fordel Studios

Feb 2025 – Present

- Implemented and integrated Golang REST APIs for production web and mobile applications.
- Integrated third-party services including payment gateways and verification APIs.
- Optimized PostgreSQL queries through indexing and schema improvements.
- Contributed to Flutter and React feature development in Agile sprints.

Full Stack Developer Intern — Fordel Studios

Aug 2024 – Jan 2025

- Refactored backend APIs and improved database query structures.
- Built additional endpoints supporting new frontend features.
- Assisted in debugging and maintaining production issues.

Key Projects

Dooars Dil — Location-Based Dating Application (Personal)

- Developed backend services for user authentication, profile management, and location-based matching in Go.
- **Challenge:** Handling hyper-local matchmaking efficiently, filtering active users in real-time, and supporting low-latency voice-first interactions.
- **Solution:** Designed a highly optimized relational schema with PostGIS for specific town clusters instead of expensive generic radius queries.
- **Implementation:** Built a concurrent voice-first messaging architecture supporting real-time voice notes, utilizing goroutines for async audio processing and efficient database connection pooling.
- Stack: Go, PostgreSQL, PostGIS, REST APIs

ZenliAI — AI Content Platform (Freelance)

- Architected a scalable Go backend using clean architecture principles with dependency injection and repository patterns.
- **Challenge:** Managing long-running generative AI API calls, merging multi-modal responses, generating complex PDFs, and tracking user token credits accurately without blocking UI threads.
- **Solution:** Implemented a robust asynchronous job queue and worker pool using Redis and Asynq for non-blocking content generation and background task management.
- **Implementation:** Integrated transactional database operations to guarantee ACID compliance during credit deductions and handled WebSocket events for real-time progress updates to the React frontend.
- Stack: Go, PostgreSQL, Redis (Asynq queue), Docker, WebSockets

Vayuroom — WebRTC Platform (Personal)

- Developed peer-to-peer WebRTC video conferencing system with Firebase as the ephemeral signaling server.
- **Challenge:** Establishing secure chat and video call encryption across multiple peers without routing traffic through a central vulnerable media server.
- **Solution:** Implemented a zero-knowledge, end-to-end encrypted architecture using WebRTC Data Channels and DTLS-SRTP protocols.
- **Implementation:** Designed a custom mesh-networking routing algorithm for small-group P2P calls, ensuring cryptographic keys are exchanged out-of-band and never stored persistently.
- Stack: React, WebRTC, Firebase, Cryptography (E2EE), Mesh Networking

AI Conversation & Sales Intelligence Platform (MVP)

- Architected a scalable Go backend using strict Clean Architecture to process real-time conversations, integrating the Gemini API for semantic intent analysis and Agent Assist suggestions. Orchestrated a Rules Engine for LLM response validation and utilized ChromaDB vector storage for reliable product knowledge retrieval, containerized entirely via Docker.
- Stack: Go, PostgreSQL, ChromaDB, Gemini AI API, Docker, Clean Architecture

Technical Skills

Backend: Go (goroutines, channels), REST APIs, JWT, Redis, PostgreSQL (schema design, indexing)

Familiar With: React, Next.js, TypeScript, Flutter, Swift

DevOps: Docker, Firebase, AWS (basic), Git

Concepts: Clean Architecture, Concurrency, API Integration, Database Design

Education

University of North Bengal — BCA

Expected May 2026